I. Engine System Overview
    a. How do oxidizer temps correlate to outside temp
        i. Fill tank will be hotter, main tank will be 10-15 degrees under outside temp
        ii. Lower than 30 degrees would mean using some sort of bottle heater
    b. Would ambient temp change vent time while sitting on pad
        i. Yes but not significantly
    c. What is the cutoff metric
        i. 5 thrust to weight is the hard cutoff
        ii. Thrust driven by pressure driven by temperature
    d. Why does thrust drop off
        i. Drain of liquid
        ii. Hard drop off at gas phase
    e. How long will the tank actually tank to drain
        i. 30 minutes + wait 30 extra to be safe
    f. Will we use a Nitrous Oxide detector to approach rocket
        i. No due to cost/supply issues (very expensive)
        ii. Will use if we can get our hands on one
    g. What launch rod will be used
        i. URRG experimental rail (up to 23 feet in height + extra if needed)
    h. What is stopping Ox from flowing out of the tank after QD
        i. Remote fill valve will be closed
    i. Is there concern about whiplash from QD popping
        i. Looking into using swing arms to prevent them from moving
    j. Is the max hold time on the pad
        i. One hourish, call made on fly
        ii. Driven by ox tank
    k. Do we own a spyglass capable of seeing 500ft
        i. Yes, Oscar has
    l. Is there concern about the weight of the piston and fuel
        i. No, the bulkhead is holding back up to 800psi of pressure
    m. Would draining fuel and ox onto the lauchpad in case of scrub cause issue
        i. Assuming not an issue, will look into further
    n. Is the signal that moves the servo one time or continuous
        i. Can be whatever we want, will be continuous
    o. Is the servo connection to the flight computer or direct PWM
        i. PWM
    p. Is it possible for the servo to only open the valve halfway before umbilical DC

        i. Can find out by running mock launches
        ii. Can also give umbilical some slack
    q. Are we taking fuel quality into account
        i. Using lab grade IPA and fuel orifices aren't small enough to clog with small impurities
    r. Are sensors calibrated
        i. Yes and will do again day of launch

II. Aerostructures
    a. For bulkheads 5 and 6 do we plan on using G10
        i. Yes, made of fiberglass
    b. Is the combustion happening next to the fins/sticking out
        i. Nothing sticks out except a quarter inch of the nozzle
    c. How will we verify the load of the fiberglass to the thrust
        i. Not looked into yet
    d. Why two threaded rods in av bay instead of rod
        i. Deemed better structure wise
    e. Is weight limits finalized
        i. No but will likely remain around 48lb
    f. Is this an L3 class rocket and if so are we concerned about drag
        i. Yes there is a concern about drag but no analysis or designs have been finalized
    g. How far down is the QD on the tube
        i. No final decision
    h. Is the only difference between raceway and piping the aero
        i. No, the piping would not just be exposed piping
    i. What wires go through the raceway
        i. Thermistor and pressure transducer lines
    j. Have we considered making rocket diameter 5" and routing through the body
        i. Yes but design already accounts for current diameter and is more expensive
        ii. Makes other things like vent tube more complicated
    k. Any concerns with av with wires being as long as they are
        i. No, they are already commonly this long
    l. What is max feed line pressure
        i. 850psi cutoff
    m. Will the raceway be inline with the fins
        i. No that is just how its currently represented in CAD
    n. What type of tube will be used =

        i. All lines are currently copper
        ii. May switch to stainless steel in the future but that will make other things more complicated
    o. What properties would we like the liner to have
        i. Low thermal conductivity
    p. What temps are we predicting for outer CC wall
        i. 240 Celsius after 5s burn, drops to 160 after 30 seconds
        ii. Unsure for how long that temp would last for with everything integrated
    q. Consider using cork for liner
    r. Would it be advantageous to use a metal fin instead of composite
        i. Not looked into, if anything it would be worse heat wise
    s. Is there a window to look at to design fins
        i. Will look into variables that will impact fin design
    t. Actual rod length
        i. Analysis at 20ft might be closer to 18-19ft

III. Avionics
    a. What keeps the sled together
        i. Endcaps slot into sleds and screw into place
    b. Do we have alternatives for JSTM connectors
        i. XT30s
    c. What is the budget for this rocket
        i. Roughly $5000 including engine development, flight vehicle is around $2000
    d. Do trackers need to be COTS
        i. Likely does not include trackers
    e. Make sure to have backups for trackers at launch
    f. Why is there a COTS flight computer requirement
        i. Tripoli requirement
    g. Is there concern for radio transparency/placement
        i. Not really the av bay should be far enough away from the tank to cause major issues
    h. How many pressure transducers can the DAQ support
        i. 2
    i. Are we actually using different connectors in the different boards
        i. Might switch to use only XT30s to be on the safe side due to recent issues

j. How will we address the BLT battery span (800miliamp battery lasted only 1 hour)
    i. Will look into scaling up battery
k. Is there plans to get power from generator to av bay
    i. Generator mainly for ground support equipment, no plans to power anything in the rocket with generator
l. Internal cam in the feeds by would be helpful
m. Will we have an audio cue for audio activated camera
    i. We can just put the buzzer/light rail from the tests on the launch
n. Is there a way to strengthen the wire with zip ties or something similar
    i. Yes we could set in epoxy or something similar

IV. Ground Support Equipment
a. Is the ethernet cable too long
    i. Looking into using a network switch to boost
b. Equipment list
    i. We have and will be implemented in future SOPs
c. What happens to QD if loss of power
    i. They will stay on
d. Will the QD popping be enough to swing the arms
    i. Maybe, other options to make sure the arm actually swings
e. What is our effective rod length
    i. Unknown as of now, haven't decided on rail button locations
f. Is a successful QD disconnect a gateway for ignition
    i. Yes, if it fails it is grounds for a scrub
g. How tall is the rocket
    i. Roughly 10 feet
h. Are we concerned with the fuel fill tower blowing over
    i. Not really but we can use sandbags or something similar to reinforce
i. Is there concern for wind blowing the bucket
    i. Once its up the rope will be pretty short so it shouldn't swing very much
j. What would we do if the ground station didn't exist
    i. Use a laptop